Powerful solutions of $\sigma(n) = 2\sigma^*(n)$: a structural reduction of an OEIS conjecture

Working notes

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Abstract

Let σ be the sum-of-divisors function and σ^* the sum-of-unitary-divisors function. We study the conjecture (R. Stephan, "Prove or disprove" #35; OEIS A063880) that

$$\sigma(n) = 2\sigma^*(n) \implies n \equiv 108 \pmod{216}.$$

After reducing to a statement about *powerful* numbers, we record a body of results that completely settle several sub-cases and isolate the remaining difficulty to a single precise question. The centrepiece is an exact identity, $R(2^2)R(3^3) = 2$, which collapses the conjecture to two local conditions on the prime factorisation. We prove the two-prime case outright, settle the $6 \mid n$ case up to one explicit residual equation, and establish 2-adic, cyclotomic, and abundancy constraints. We then locate the open kernel exactly: it is the inability to bound $\omega(n)$, the same obstruction that has kept unitary perfect numbers open since 1966 — though, unlike those problems, ours carries a *convergent* heuristic predicting finiteness.

1 Setup

For $n = \prod_i p_i^{a_i}$ write $\sigma(n) = \prod_i \frac{p_i^{a_i+1}-1}{p_i-1}$ and $\sigma^*(n) = \prod_i \left(p_i^{a_i}+1\right)$, the latter being the sum of the unitary divisors $d \parallel n$ (those with $\gcd(d, n/d) = 1$). Both are multiplicative. Define

$$R(n) = \frac{\sigma(n)}{\sigma^*(n)} = \prod_{p^a \parallel n} R(p^a), \qquad R(p^a) = \frac{1 + p + \cdots + p^a}{p^a + 1} = \frac{p^{a+1} - 1}{(p-1)(p^a+1)}.$$

The defining equation is exactly $R(n) = 2$. The following elementary facts are used throughout.

Lemma 1. *For a prime p and $a \geq 1$:*

(i) $R(p) = 1$*;*

(ii) $R(p^{a+1}) - R(p^a) = \dfrac{p^a(p+1)}{(p^a+1)(p^{a+1}+1)} > 0$*, so $R(p^a)$ is strictly increasing in a;*

(iii) $R(p^a) < \dfrac{p}{p-1}$*, with $R(p^a) \uparrow \dfrac{p}{p-1}$ as $a \to \infty$;*

(iv) consequently $R(p^a) > 1$ for all $a \geq 2$, and $R(n) = R(m)$ where m is the powerful part *of n (the product of the prime powers $p^a \parallel n$ with $a \geq 2$).*

By (i) and (iv), squarefree prime factors are invisible to R. Hence every solution n factors as $n = m \cdot s$ with m powerful, s squarefree, $\gcd(m, s) = 1$, and $R(n) = R(m)$. The conjecture therefore reduces to a statement about the powerful "cores".

Crux. *The only powerful m with $R(m) = 2$ is $m = 108 = 2^2 \cdot 3^3$.*

If the Crux holds, then every solution is $108\,s$ with s squarefree and coprime to 6, whence $n \equiv 108 \pmod{216}$, which is the conjecture. (This reduction, and a verification of the Crux to 10^{18}, were also obtained independently by A. Eldar in the OEIS entry; the Crux itself remains open.)

2 The collapse

Theorem 2 (Collapse / reduction)**.** *$R(2^2)\,R(3^3) = \frac{7}{5} \cdot \frac{10}{7} = 2$. Consequently, if a powerful solution m satisfies $4 \parallel m$ and $27 \parallel m$, then $m = 108$. Equivalently,*

$$\text{the Crux} \iff \text{every powerful solution } m \text{ has } v_2(m) = 2 \text{ and } v_3(m) = 3.$$

Proof. Suppose $4 \parallel m$, $27 \parallel m$, and write $m = 2^2 \cdot 3^3 \cdot M$ with M powerful and coprime to 6. Then $R(m) = R(2^2)R(3^3)R(M) = 2R(M)$, so $R(m) = 2$ forces $R(M) = 1$. By Lemma 1(iv), $R(M) > 1$ for any powerful $M > 1$; hence $M = 1$ and $m = 108$. □

Thus $2^2 \cdot 3^3$ *exactly exhausts* the budget $R = 2$: any further powerful factor strictly increases R. This is the structural reason 108 is the answer, and it explains the modulus $216 = 2^3 \cdot 3^3$ in the conjecture ($v_2 = 2$ together with $27 \mid n$).

3 The two-prime case

Theorem 3. *The only powerful m with $\omega(m) \leq 2$ and $R(m) = 2$ is 108.*

Proof. If $\omega(m) = 1$ then $R(p^a) < \frac{p}{p-1} \leq 2$ by Lemma 1(iii), strictly, so no solution. For $\omega(m) = 2$, two *odd* primes $3 \leq P < Q$ give $R(m) < \frac{P}{P-1} \cdot \frac{Q}{Q-1} \leq \frac{3}{2} \cdot \frac{5}{4} = \frac{15}{8} < 2$; so a two-prime solution has the form $m = 2^a P^b$ with P odd, $a, b \geq 2$.

Use the identity $\sigma(2^a) = 2^{a+1} - 1 = 2(2^a + 1) - 3 = 2\sigma^*(2^a) - 3$. Writing $u = 2^a + 1$, the equation $\sigma(2^a)\sigma(P^b) = 2u(P^b + 1)$ becomes

$$2u\big(\sigma(P^b) - P^b - 1\big) = 3\,\sigma(P^b), \qquad \text{i.e.} \qquad 2u\,(P + P^2 + \cdots + P^{b-1}) = 3\,\sigma(P^b).$$

Reducing mod P: the left side is $\equiv 0$ (each term carries a factor P, as $b \geq 2$), while $\sigma(P^b) \equiv 1$. Hence $0 \equiv 3 \pmod{P}$, so $P = 3$. With $P = 3$ the equation reduces (set $y = 3^{b-1}$) to $u = \frac{9y-1}{2(y-1)} = \frac{9}{2} + \frac{4}{y-1}$; for $u = 2^a + 1$ a positive integer this forces $y = 9$, i.e. $b = 3$, $a = 2$. Thus $m = 2^2 \cdot 3^3 = 108$. □

4 The case $6 \mid m$

Theorem 4. *Let m be a powerful solution with $6 \mid m$. Then $v_2(m) = 2$ and $v_3(m) \in \{2, 3\}$. Moreover:*

- *if $v_3(m) = 3$ then $m = 108$;*
- *if $v_3(m) = 2$ then $m = 36\,M$ with M powerful, coprime to 6, and $R(M) = \frac{100}{91}$; every such M has $\omega(M) \geq 3$ and all prime factors ≥ 11.*

In particular, $6 \mid m$ implies $m = 108$ unless the residual equation $R(M) = \frac{100}{91}$ has a solution.

Proof. Write $m = 2^a 3^b M$, $a, b \geq 2$, M powerful coprime to 6, $R(M) \geq 1$. If $a \geq 3$ then $R(m) \geq R(2^3)R(3^2) = \frac{5}{3} \cdot \frac{13}{10} = \frac{13}{6} > 2$, impossible; so $a = 2$ and $R(3^b)R(M) = \frac{2}{R(2^2)} = \frac{10}{7}$. If $b \geq 4$ then $R(3^b) \geq R(3^4) = \frac{121}{82} > \frac{10}{7}$, impossible; so $b \in \{2, 3\}$. For $b = 3$, $R(M) = \frac{10/7}{10/7} = 1$, so $M = 1$ and $m = 108$. For $b = 2$, $R(M) = \frac{10/7}{13/10} = \frac{100}{91}$.

For the residual: if a prime $p \leq 7$ divided M then $R(M) \geq R(p^2) \geq R(7^2) = \frac{57}{50} > \frac{100}{91}$, impossible; so every prime of M is ≥ 11. A single prime power cannot equal $\frac{100}{91}$ (for $p = 11$ the value $\frac{100}{91}$ lies strictly between $R(11^2) = \frac{133}{122}$ and $R(11^3) = \frac{122}{111}$; for $p \geq 13$, $R(p^a) < \frac{p}{p-1} \leq \frac{13}{12} < \frac{100}{91}$), so $\omega(M) \geq 2$. The case $\omega(M) = 2$ is excluded by a finite computation (the two-sided bound $R(p^a) \in [R(p^2), \frac{p}{p-1})$ confines the second prime to an interval that contains no admissible value), leaving $\omega(M) \geq 3$. $\square$

Remark 5. The residual is also empty below $\sim 10^{16}$: if $R(M) = \frac{100}{91}$ then $R(36M) = 2$, so $36M$ would be a powerful solution other than 108, contradicting the verification to 10^{18}.

5 Structural constraints

Theorem 6 (2-adic identity). *Let $m = 2^{e_0} \prod_{i=1}^{k} P_i^{e_i}$ with the P_i odd, and let $O = \{i : e_i \text{ odd}\}$. Any solution satisfies*

$$\sum_{i \in O} v_2(e_i + 1) = 1 + k.$$

In particular $O \neq \varnothing$, and at least one odd prime occurs to an exponent $\equiv 3 \pmod 4$.

Proof sketch. By the lifting-the-exponent lemma at 2, for odd P one has $v_2(\sigma(P^e)) = v_2(P+1) + v_2(e+1) - 1$ when e is odd and 0 when e is even, while $v_2(P^e + 1) = v_2(P+1)$ or 1 accordingly. Matching v_2 across $\sigma(m) = 2\sigma^*(m)$, the terms $\sum_{i \in O} v_2(P_i + 1)$ cancel, leaving the stated identity. For 108: $k = 1$, $O = \{3\}$, and $v_2(3+1) = 2 = 1 + 1$. $\square$

Theorem 7 (Cyclotomic entanglement). *For each prime power $p_i^{a_i} \parallel m$ (with the sole exception 2^5), the primitive prime divisor q of $\Phi_{a_i+1}(p_i)$ — which satisfies $q \equiv 1 \pmod{a_i + 1}$ — divides $p_j^{a_j} + 1$ for some $j \neq i$.*

Proof sketch. By Zsygmondy's theorem such a q exists ($a_i + 1 \geq 3$), divides $\sigma(p_i^{a_i})$, hence the right-hand side $2\prod_j (p_j^{a_j} + 1)$. As q is odd it divides some $p_j^{a_j} + 1$; and $j = i$ is impossible, since $q \mid p_i^{a_i} + 1$ would force $(a_i + 1) \mid 2a_i$, i.e. $a_i \leq 1$, contradicting powerfulness. $\square$

Proposition 8 (Abundancy lower bounds). *The bound $\prod_{p \mid m} \frac{p}{p-1} > 2$ holds for every solution. Hence if m is coprime to 6 then $\omega(m) \geq 7$, and the bound grows sharply as small primes are excluded:*

least prime of m	5	7	11	13	17	19	23
$\omega(m) \geq$	7	15	27	41	62	85	115

6 Computational and heuristic status

Verification. The only powerful solution below 10^{18} is 108; equivalently, no solution $n \leq 10^{18}$ violates the conjecture.

Why finiteness is expected. Powerful numbers have Dirichlet series $\sum_{m \text{ powerful}} m^{-s} = \zeta(2s)\zeta(3s)/\zeta(6s)$, so

$$\sum_{m \text{ powerful}} \frac{1}{m} = \frac{\zeta(2)\zeta(3)}{\zeta(6)} \approx 1.944 < \infty,$$

and their counting function is $\sim \frac{\zeta(3/2)}{\zeta(3)}\sqrt{x} \approx 2.173\sqrt{x}$. Modelling $\sigma(m) = 2\sigma^*(m)$ as a value hit with probability $\asymp 1/m$ gives a *finite* expected number of powerful solutions, with tail beyond X of size $\asymp X^{-1/2}$ (about 2×10^{-9} beyond 10^{18}). This convergence — in contrast to the logarithmically divergent sums underlying perfect and unitary perfect numbers — is exactly what powerfulness buys, and is the strongest reason to believe 108 is the whole answer.

What rigour gives, and where it stops. Dickson's 1913 method (finitely many odd perfect numbers with at most k prime factors) carries over verbatim:

Theorem 9 (Dickson-type finiteness)**.** *For each fixed k, there are effectively finitely many powerful m with $\omega(m) = k$ and $R(m) = 2$.*

(If $\omega(m) = k$ then $\left(\frac{p_1}{p_1-1}\right)^k > 2$ bounds the least prime; recursion bounds all primes, and an S-unit / linear-forms-in-logarithms argument then bounds the exponents.) Consequently **the Crux is decidable as soon as ω is bounded.** Every result above is consistent with this; the single missing ingredient is an upper bound on $\omega(m)$.

That bound is exactly the open obstruction. As we showed, the two-sided window $\prod_{p|m} R(p^2) \leq 2 < \prod_{p|m} \frac{p}{p-1}$ cannot cap ω: a single small prime can satisfy the upper inequality, freeing the remaining primes to be arbitrarily large. This is the same wall that leaves *unitary perfect numbers* open (Subbarao's finiteness conjecture, 1966), and that places the value $2 = 2/1$ in the hard regime — in the abundancy-index analogue $\sigma(n)/n = r$, finiteness is known to be generically intractable (Pomerance's $r = 93/40$, with solutions exactly $\{80, 200\}$, is essentially the only nontrivial r for which finiteness with more than one solution has been proved, and it succeeds only because of its constraining denominator).

7 Summary

Proven shell. The two-prime case (Theorem 3); the collapse reducing the Crux to "$v_2 = 2$ and $v_3 = 3$" (Theorem 2); the case $6 \mid m$ up to the residual $R(M) = \frac{100}{91}$, itself resolved for $\omega(M) \leq 2$ and verified empty below $\sim 10^{16}$ (Theorem 4); the 2-adic identity (Theorem 6); cyclotomic entanglement (Theorem 7); and abundancy bounds (Proposition 8).

Open kernel. Powerful solutions with $6 \nmid m$ (forced to have ≥ 7 prime factors if coprime to 6), together with the $\omega \geq 3$ residual. This kernel is of odd-perfect / unitary-perfect grade: the entire gap is a bound on $\omega(m)$, which no current method supplies.

Belief. Theorem 9 plus the convergent heuristic make finiteness — and indeed uniqueness of 108 — overwhelmingly likely; what is missing is a proof, not evidence.

References (informal). R. Stephan, *Prove or disprove. 100 conjectures from the OEIS*, arXiv:math/0409509, #35; OEIS A063880 (and comments by A. Eldar); L. E. Dickson, *Finiteness of the odd perfect and primitive abundant numbers with n distinct prime factors*, Amer. J. Math. 35 (1913); C. R. Wall, *Bi-unitary perfect numbers*, Proc. AMS 33 (1972) [bi-unitary perfect = {6, 60, 90}]; M. V. Subbarao & L. J. Warren, *Unitary perfect numbers*, Canad. Math. Bull. 9 (1966); P. Pollack & C. Pomerance,

Some problems of Erdős on the sum-of-divisors function, and related work on the distribution of $\sigma(n)/n$.